

NanoGT Match Report: Fragile X syndrome

Disease: Fragile X syndrome (ORPHA:908)

Primary gene: FMR1

Gene CDS: 1899 bp

Inheritance: X-linked dominant

Target tissues scored: CNS

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Vector does not naturally cover all annotated disease tissues: cns; Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility; Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable.

Disease Mechanism Evidence

Molecular mechanism: repeat expansion silencing

Mechanistic detail: CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed

Gene-addition compatibility: conditional

Preferred modality class: aav cns fmr1 cdna addition

Evidence level/status: direct / source_checked

Evidence summary: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia

Evidence source: Liu XS et al. 2018 Cell PMID 29456084

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
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Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Skysona	LV	7.0/10	Medium	approved
2	Libmeldy	LV	6.8/10	Medium	approved
3	OAV101-IT	AAV9	6.8/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
4	Zolgensma	AAV9	6.8/10	Medium	approved
5	Strimvelis	LV	6.7/10	Medium	approved

Match #1: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 7.0 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.95	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1899bp / cargo 8000bp (24% utilized)
- Precedent target match: cns
- Protein class mismatch

- Inheritance match (X-linked dominant <-> XL)
- Unknown pathway — neutral score
- Disease mechanism: repeat expansion silencing — CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia
- Mechanism source: Liu XS et al. 2018 Cell PMID 29456084 (<https://pubmed.ncbi.nlm.nih.gov/29456084/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — FMR1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity

Dimension	Score	Max	What it measures
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.81	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1899bp / cargo 8000bp (24% utilized)
- Precedent target match: cns
- Protein class mismatch
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: repeat expansion silencing — CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia
- Mechanism source: Liu XS et al. 2018 Cell PMID 29456084 (<https://pubmed.ncbi.nlm.nih.gov/29456084/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — FMR1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: cns
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #3: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.81	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1899bp / cargo 4700bp (40% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: repeat expansion silencing — CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia
- Mechanism source: Liu XS et al. 2018 Cell PMID 29456084 (<https://pubmed.ncbi.nlm.nih.gov/29456084/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — FMR1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #4: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.81	10.0	Raw sum / 21 × 10 (v2: 14 dimensions)

Rationale

- Gene CDS 1899bp / cargo 4700bp (40% utilized)
- Vector tropism plus precedent target match: cns
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: repeat expansion silencing — CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia
- Mechanism source: Liu XS et al. 2018 Cell PMID 29456084 (<https://pubmed.ncbi.nlm.nih.gov/29456084/>)
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — FMR1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	6.71	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 1899bp / cargo 8000bp (24% utilized)
- No tissue overlap (disease: ['CNS'], vector: ['hematopoietic'])
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Disease mechanism: repeat expansion silencing — CGG trinucleotide repeat expansion (>200 repeats) in the FMR1 5-prime UTR triggers de novo CpG methylation and transcriptional silencing; the FMR1 protein coding sequence is structurally normal when expressed
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Liu XS et al. (2018 Cell PMID 29456084) demonstrated methylation-dependent silencing of the FMR1 locus by the CGG repeat; FMR1 cDNA addition (lacking the expanded repeat-containing UTR) bypasses the silenced endogenous allele and is not subject to re-silencing — same logic as FXN in Friedreich ataxia

- Mechanism source: Liu XS et al. 2018 Cell PMID 29456084 (<https://pubmed.ncbi.nlm.nih.gov/29456084/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — FMR1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Vector does not naturally cover all annotated disease tissues: *cns*
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition